SS 700:2023
(ICS 97.220.10)

SINGAPORE STANDARD

Code of practice for water safety – Aquatic facilities

Singapore Standards Council

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Foreword

This Singapore Standard was prepared by the Working Group on Water Safety in Swimming Pools set up by the Technical Committee on Workplace Safety and Health under the purview of the Safety and Quality Standards Committee.

This Singapore Standard was developed as a result of the review of TR 71:2019 Code of practice for water safety – Aquatic facilities. This standard contains additional information for the handling of shallow water blackouts and prolonged breath-holding for the persons who work at and/or patronise aquatic facilities.

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2. Kent County Council

   – Sections 1, 2 and 3 of Annex B from Emergency action plan for Headcorn Primary School

3. Royal Life Saving Society Australia

   – Clause 9 from Operations manuals, Guidelines for safe pool operation GO1

   – Clause 16 from Hire of facilities, Guidelines for safe pool operation GO4

4. Sport Singapore

   – Annex A from Preliminary – Major accident report and major/minor incident report
   – Annex G from Safety management's water safety outreach materials
   – Figure C.1 and Tables C.1 to C.4 from Risk management in sport

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Code of practice for water safety – Aquatic facilities

1 Scope

This standard specifies the general requirements aimed at preventing drowning and injuries at aquatic facilities.

2 Normative references

The following referenced documents are indispensable for the application of this standard. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

SS 556	Code of practice for the design and management of aquatic facilities
SS 586-2	Specification for hazard communication for hazardous chemicals and dangerous goods − Part 2: Globally harmonised system of classification and labelling of chemicals – Singapore's adaptations
SS 681:2022	Code of practice for sport safety

3 Terms and definitions

For the purpose of this Singapore Standard, the following terms and definitions apply.

3.1 Aquatic facility

A man-made body of water used for sport, recreation and/or therapeutic water activities.

3.2 Automated external defibrillator (AED)

A portable electronic device that automatically diagnoses the life-threatening cardiac arrhythmias of ventricular fibrillation and ventricular tachycardia (abnormally rapid heart rate) in a patient and is able to treat them through defibrillation.

Note 1 to entry: Cardiac arrhythmia is a condition in which the heart beats with an irregular or abnormal rhythm.

3.3 Cardiopulmonary resuscitation (CPR)

An emergency procedure that combines chest compressions often with artificial ventilation in an effort to manually preserve intact brain function until further measures are taken to restore spontaneous blood circulation and breathing in a person who is in cardiac arrest.

3.4 Drowning

A process of experiencing respiratory impairment from submersion/immersion in liquid.

Note 1 to entry: Drowning outcomes are classified as death, morbidity and no morbidity.

[SOURCE: World Health Organisation]

3.5 Emergency action plan (EAP)

A pre-determined, documented and rehearsed plan of action implemented upon the occurrence of an emergency.

3.6 Feature pool

A pool equipped with facilities such as lazy river, spa, water slides and/or wave pool.

3.7 Hazard

Anything with the potential to cause bodily injury, including any physical, chemical, biological, mechanical, electrical or ergonomic hazard.

3.8 Hydrotherapy/physiotherapy pool

A pool containing heated or unheated water, designed to meet the therapeutic needs of persons of any age with impairments due to illness, injury, disease, intellectual handicap, and/or congenital defects. It can also be used for fitness, recreational and educational purposes. The pool can also incorporate, or is connected to, equipment for heating the water contained in it and injecting air bubbles or jets of water under pressure to create general turbulence in the water.

3.9 Intentional hyperventilation

Intentional hyperventilation seeks to reduce carbon dioxide (CO_2) in the blood by rapid inhalation and exhalation, thereby delaying stimulation of the brain to breathe during prolonged breath-holding, even when blood oxygen levels are low. This places the individual at risk of losing consciousness before surfacing and drowning.

Note 1 to entry: As the breath-hold continues, oxygen (O_2) levels in the blood drop. Usually, an increase in CO_2 in the blood can trigger the desire to breathe and the swimmer can end the breath-hold attempt. Due to hyperventilation, CO_2 levels are too low to trigger this, and the urge to breathe does not occur as early. This, combined with low O_2 levels, can cause the swimmer to go unconscious. Once unconscious, the body can react by initiating a breath. However, whilst submerged, the lungs can be filled with water. If immediate assistance does not occur, death by drowning can occur.

3.10 Interactive fountain/water playground or pool

Water sprays or jet devices in a playground or pool located in an area accessible to the public. They are intended to provide individuals with a means to play in the water.

3.11 Landing pool

A body of water located at the exit of a waterslide, used to break the fall of waterslide users.

Note 1 to entry: Waterslide is a device incorporating an inclined sliding surface, where a user’s body comes into direct contact with a water medium that is used to propel, or decelerate a body within a water flume, which ends in a landing pool and/or watershed area.

3.12 Lazy river

An aquatic facility that is designed to simulate the effects of a natural river, which incorporates a system to produce an artificial current of water, created to propel users along with or without the use of a floating device/vessel.

3.13 Leisure pool

A pool used for recreational purposes.

3.14 Lifeguard

A person who is appropriately qualified and experienced, with the primary responsibility of observing the aquatic facility to anticipate problems, hazards, identify emergencies, carry out rescues and give immediate first aid.

3.15 Major incident

An incident classified as life-threatening.

3.16 Minor incident

An incident resulting in injury or damage that is not deemed to be life-threatening.

3.17 Occupier

The person who oversees management or control of a premises, whether as the owner of the premises or an agent of another person.

3.18 Personal protective equipment (PPE)

Equipment, such as clothing, worn by individuals to protect against potential hazards.

3.19 Pre-participation questionnaire

A self-administered questionnaire, such as the Get Active Questionnaire (GAQ), for participants to check on their personal physical well-being before engaging in any form of physical activity.

3.20 Prolonged breath-holding

Intentionally holding one's breath while submerged and ignoring the need to breathe, often preceded by intentional hyperventilation to extend the duration (see 3.9).

3.21 Qualified person

A person who, through formal training certification from relevant agencies, possesses the practical and theoretical knowledge of specified equipment to enable the detection of defects or weaknesses and to assess their fitness for the intended purpose and safety for continued use.

3.22 Risk

The likelihood that a hazard can cause a specific bodily injury to any person.

3.23 Shallow water blackout / Hypoxic blackout

Loss of consciousness underwater due to lack of oxygen to the brain, usually from holding one's breath for long periods of time.

3.24 Spa pool

A man-made pool or other water-retaining structure designed for recreational activities other than swimming, which can be heated or unheated. It incorporates, or is connected to, equipment for heating the water contained in it and injecting air bubbles or jets of water under pressure to create general turbulence in the water.

3.25 Standard operating procedure

A procedure normally carried out on a day-to-day basis to ensure the smooth, safe, effective and efficient operation of a facility.

3.26 Starting platform

An elevated structure located on the side of a swimming pool, designed for use by swimmers in competitive lap-swimming events.

3.27 Supervision

Adequate and constant surveillance of persons in and around the water by a person appropriately qualified to anticipate problems, identify emergencies quickly and provide an appropriate and timely response (preferably but not limited to lifeguards).

Supervision by swim instructors and/or caregivers at all times entails watching persons, especially weak swimmers, persons with underlying medical conditions and/or persons who are physically challenged.

3.28 Swimming pool

A man-made structure located either indoors or outdoors, containing treated water, heated or unheated, intended for swimming and other recreational purposes. This definition does not include spa pools, individual therapeutic tubs or baths used for cleaning of the body.

3.29 Training

Education, instruction and practice that enable individuals to demonstrate competency.

3.30 Turnover period

A change of water within a water pool (i.e. three turnover periods indicates that the water within a water pool is changed three times).

3.31 Wading pool

A swimming pool designed for wading, where the water depth is typically less than 500 mm.

3.32 Wave pool

An aquatic facility designed to simulate the effects of a beach and incorporates a system to produce artificial wave motion.

3.33 Use of “shall”, “should”, “may” and “can”

In this standard, the following verbal forms are used:

– “shall” indicates that the requirement is strictly to be followed in order to conform to the standard and from which no deviation is permitted;

– “should” indicates a recommendation;

– “may” indicates a permission; and

– “can” indicates a possibility or a capacity.

4 Requirements for aquatic facility management

4.1 Classification of aquatic facilities

For the purposes of this standard, aquatic facilities are classified in Table 1.

Table 1 – Classification of aquatic facilities

Classification	Examples
Man-made water bodies used for sports and recreation purposes.	Swimming pools, leisure pools, wave pools, wading pools, diving pools, landing pools, water slides and lazy rivers, etc.
Man-made water bodies that are designed for activities (other than sports) and are generally aerated and can be heated or not heated.	Spa pools and hydrotherapy/physiotherapy pools, etc.
Man-made water features that are designed for interactive play in an area accessible to the public.	Interactive fountain/water playground, etc.

4.2 Roles and responsibilities

4.2.1 Occupier

The occupier of any aquatic facility shall be responsible for the safety of the premises including the common public access area, pool infrastructure and equipment for all pool users and employees.

Where the pool is an ancillary part of a larger complex (e.g. a hotel, hospital, condominium or school), the facility occupier should appoint a lifeguard or install a monitoring system such as CCTV cameras to monitor the pool safety. The facility occupier shall also provide lifesaving equipment readily available for use in the event of an emergency. There shall be outreach and educational efforts on the use of such lifesaving equipment (e.g. through demonstrations, signages, posters, etc.).

If there are activities organised beside or near the pool, the occupier should also appoint either a lifeguard, spotter or volunteer to keep a lookout for persons in distress and provide immediate assistance. An emergency contact list shall be made visible and displayed, including the name and contact details of the duty personnel.

The occupier shall conduct a risk assessment that minimally includes the following:

a) Weather

b) Facility

- Size, number and layout of pools;
- Remoteness/location;
- Lighting and surface reflection.

c) Number and distribution of users

- Peak periods and holidays;
- Average attendance.

d) Profile of swimmers

- Swimming capabilities;
- Special needs individuals and groups; and

e) Activities (either programmed or spontaneous)

- Recreational activities;
- Competitions or pool parties.

The responsibilities of the occupier should also include, but are not limited to the following:

a) Ensuring adequate insurance coverage for any injuries within the pool facility and damage to property.

b) Pro-actively promoting lifesaving, CPR and AED skills regularly through educational workshops, training courses and other appropriate means.

c) Promoting the recommendations stipulated in the Water Safety Code (see 7.2) to all relevant stakeholders.

d) Documenting the maintenance and incident records (major and minor) for review and improvement (see Annex A).

e) Investigating any drowning cases, injuries (major or minor) and/or near misses. Any investigation should identify root causes and contributing causes for rectification, learning and improvement. Briefing and/or training of the revised procedures should be conducted immediately for staff and management of the facility.

f) Sharing of incident findings and statistics with the management team at least annually, and

g) Monitoring the number of users within the facility at any one time to determine supervision requirements. A large increase in the number of users in and/or around water bodies can lead to an increase in risk and, hence, a need for increased supervision.

4.2.2 Lifeguard

A lifeguard shall educate and enforce the rules and regulations of aquatic facilities, encourage safe behaviours, manage and report incidents. They should proactively promote awareness of specific and general hazards within the facility.

4.3 Information at aquatic facilities

4.3.1 General

The following information and advisory signs shall be provided at aquatic facilities:

a) An EAP or procedures that relate to the specific pool or pools, detailing specific instructions on the action to be taken by staff;

b) An emergency support system with an effective means of communication that shall include a public address system, siren or loud hailers that can be used for announcements and a direct telephone link to an appropriate emergency service;

c) First aid, CPR and AED posters; and

d) Rules and regulations such as appropriate and acceptable behaviours for pool users, supervision of children, dangers of underwater breadth holding.

4.3.2 Emergency action plan (EAP)

The EAP shall be regularly reviewed and updated, especially after the installation of new play equipment, a major renovation and/or change in staff. All revisions shall be made aware to all staff. A detailed checklist outlining key features of both the operations manual and the EAP shall be included (see Annex B for a sample of EAP).

The EAP shall include safe and timely access by first responders in an emergency to improve the chance of survival of the collapsed person. For example, climbing over a locked toilet cubicle to access the collapsed person.

Other potential emergencies can also arise around the pool deck, toilets (e.g. within locked cubicles), offices, etc. Appropriate rescue procedures should be developed, communicated and put into practice.

The EAP and layout plan of the facility shall be provided to the civil defence authority to aid their emergency and rescue efforts. The layout plan shall also highlight high-risk areas, such as the plant room and the chemicals used, as this can enable the civil defence personnel to approach the emergency in a more strategic and efficient manner.

4.3.3 Swim attire for infants and toddlers

Occupiers shall have policies in place that require infants and toddlers who are not toilet-trained to wear a fitted and approved swimming nappy for public health reasons. Any child having bowel movement shall leave or be removed immediately from the water for necessary clean up. Soiled clothing and nappies (including aqua nappies) shall be disposed of carefully. Facility personnel shall be notified immediately of any fecal content in the pool.

4.3.4 Safety policy

Safety policies for swimmers, such as wearing appropriate swim attire, refraining from engaging in unsafe behaviours, awareness of pre-participation questionnaires (e.g. GAQ) and not swimming if unwell and/or infected with a contagious disease, shall be in place.

NOTE − Refer to Sport Singapore website for information on the pre-participation questionnaire.

4.4 Safety and rescue equipment

4.4.1 General

All rescue equipment, such as a rescue tube, rope, reach pole, spinal board, resuscitation pocket mask and walkie-talkie, shall be placed in a prominent area with easy access. All rescue equipment shall be checked regularly to ensure they remain in good operational condition. Where applicable, there should be signage with pictorial instructions to guide users on their usage.

All facility staff should be trained in the use of these safety and rescue equipment prior to deployment of duty.

4.4.2 First-aid kit and equipment

A first-aid kit, AED and appropriate oxygen equipment should be made available and easily accessible. An absorbent towel should be kept in a glass box with the AED to facilitate drying of the chest before application of the AED.

Other equipment used in an emergency, such as loud hailers, public address system and/or telephones, should be made available.

4.4.3 Users who require additional assistance

Staff should receive training in managing users with disabilities, including:

a) safe lifting techniques for the emergency removal of users; and

b) appropriate communication to minimise confusion and potential embarrassment for both parties.

5 Risk management

5.1 General

Risk management is the process of identifying, assessing and controlling risks to an organisation, people, infrastructure or assets. It is an essential tool and relevant to many facets of the aquatics industry such as supervision, programming and plant room operation.

A risk assessment, pertaining to the facility and its activities, for example, pool parties, aqua aerobics, etc., shall be conducted by the facility owner, occupier, hirer and programme operator. Parents who bring their family members or guests to the aquatic facility should also look out for dangers and hazards such as inclement weather, young children and weak swimmers, persons with underlying medical conditions, and depth of water bodies. See Annex C for risk assessment.

5.2 Employers

Employers shall undertake reasonable health and safety measures towards anyone (e.g. employees, contractors, guests, etc.) entering the facility either for the conduct of business or pleasure.

5.3 Employees

Employees shall:

a) take reasonable care of their own safety and the safety of others affected by their actions or omissions;

b) co-operate with the employer with respect to any actions taken by the employer in safety and health matters; and

c) promptly inform their supervisor if they are unable to perform their assigned role effectively due to conditions such as mental or physical health issues, pregnancy, etc. In such cases, a medical document/certificate shall be submitted to the employer.

5.4 Occupier

The occupier shall provide and maintain a safe facility, including those in connection with the use, handling, storage and transfer of chemicals/substances. The occupier shall ensure that a competent person oversees a risk assessment of the aquatic facility and its activities.

The occupier shall fulfill their duty of care by ensuring the following:

a) Providing and maintaining systems of work, including those in connection with the use, handling, storage and transfer of chemicals/substances in accordance with SS 586-2.

b) Providing information, instructions, training and supervision to employees on commencement of employment, in relation to any workplace procedures and hazards. Records of the instructions, training and supervision provided shall also be kept.

c) Safekeeping records relating to the training, safety and health of employees, and

d) All the employees working in a hazardous environment are fully aware of the risks and, where applicable, to provide the necessary PPE.

5.5 Facility staff

Facility staff shall conduct a risk assessment of the aquatic facility, especially in the areas of:

a) supervision of users;

b) storage, handling methods and procedures for all hazardous chemicals; and

c) activities, equipment, manual handling, confined spaces, working at heights and ensuring hirers' stated activities on the letter of agreement are compliant with the facility's risk management plan.

5.6 Hazards

The occupier should consider the following hazards in relation to the use of the aquatic facility (see Annex D for types and examples of hazards):

a) Drowning (see Annex E for tips to prevent drowning);

b) Chemical exposure;

c) Small toys;

d) Inflatable floats;

e) Slip, trip and fall (see Annex F for prevention tips);

f) Diving;

g) Lighting;

h) Entrapment;

i) Lightning;

j) Blind spots;

k) Sharp edges from broken tiles; and/or

l) Uneven surfaces.

5.7 Records and documentation

Records of risk assessments shall be documented and regularly reviewed. It is presupposed that the risk assessment is done in accordance with applicable statutory and regulatory requirements. Risk assessments shall be reviewed immediately:

a) following a major incident;

b) when there are major changes to operation(s)/activities; or

c) when there is an introduction of regulations/guidelines/standards by regulatory authorities and/or the amendments thereof which affect the aquatic facility's daily operations.

6 Encouraging responsible behaviour

6.1 Roles of aquatic facility management and staff

The aquatic facility management and staff should proactively encourage and provide guidance on responsible behaviour. This includes educating users, parents, and caregivers on pool rules so they can assist to enforce said rules to children. For example, reminding parents to actively supervise their children, ensuring there are no glass containers in and around the pool area, and prohibiting children from running around the pool deck.

The aquatic facility management and staff should remain vigilant when there are activities in and around the pool. This includes promoting responsible behaviour among parents hosting pool parties, hiring a lifeguard for the duration of the party, or assigning a spotter or volunteer to monitor the pool to ensure that no child unintentionally falls into the pool.

Parents should be reminded of the importance of close supervision of their children, even in wading pools. They should be reminded to remain within arm's reach of their children to provide immediate assistance if needed. When parents are attentive, lifeguards can focus more on their duties.

6.2 Carry out pre-participation screening

An instructor/coach/adult supervisor shall carry out a pre-participation screening for swimmers in accordance with Clause 8 of SS 681:2022, such as administering the GAQ. They should also be advised to adopt the buddy system, i.e. swimming in pairs or with friends who are able to look out for one another.

6.3 Users' responsibility

Users of aquatic facilities shall be responsible for observing, understanding and internalising all displayed signages by facility owners or occupiers.

6.4 Shallow water blackout / Hypoxic blackout prevention

6.4.1 The average person can hold their breath for 30 to 90 s. This duration can increase or decrease due to several factors such as smoking, underlying medical conditions or breath training. A person can practise breath-holding to increase their lung capacity and there are training guidelines for individuals to learn to hold their breath for prolonged periods. This training usually takes several months.

6.4.2 The occupier should improve the supervision of swimmers by training lifeguards to enhance supervision, prohibit, recognise and respond to dangerous and/or emergent situations like hyperventilation.

6.4.3 Occupiers, lifeguards, instructors, coaches, parents and caregivers shall advise swimmers (regardless of whether they are children or adults) on the following:

a) Swimmers shall not practise prolonged breath-holding and ignore the need to breathe. If there is a need to practise, the rule of thumb for safety is one breath hold, one time only, for a maximum distance of 25 m. This activity shall not be repeated thereafter.

b) Risks increase with repetitive breath holding. If breath-holding is done underwater, a buddy shall be next to the swimmer to focus on the swimmer's safety. The buddy shall tap on the swimmer's shoulder for the swimmer to respond and signal that they are all right. The buddy shall not breath-hold together with the swimmer.

c) Swimmers shall not hyperventilate.

6.4.4 During swimming classes, coaches and instructors shall ascertain if swimmers under their care are able to progress to train for repetitive underwater lengths.

6.4.5 Coaches and instructors shall check and arrange mitigating measures for the following aspects:

a) Swimmer's respiratory issues, if any;

b) Swimmer's age; and

c) Swimmer's competency level (including swimmer's previous competency at swimming underwater).

6.4.6 Coaches/Instructors shall have relevant lifesaving skills and experience in water sports (e.g. snorkelling, scuba diving, free-diving (apnea), underwater rugby, water hockey and fin swimming) and shall have the competence to carry out their duties.

7 Water safety awareness programme

7.1 Stakeholders engagement

7.1.1 Occupier's role

The occupier should organise outreach activities for the following stakeholders:

a) Aquatic facility staff and volunteers;

b) Condominium/hotel/school management team (if there is an aquatic facility) within the premises;

c) Caregivers/guardians/parents;

d) Aquatic facility users; and/or

e) Aquatic facility event organisers.

7.1.2 Outreach programme

Outreach topics should include but are not limited to the following:

a) Pre-participation questionnaire, e.g. GAQ;

b) Stretching and warm-up exercises;

c) Shallow water blackout prevention;

d) Identification of swimmers' behaviours (see Annex G);

e) Rescue equipment usage;

f) CPR/AED efforts; and/or

g) Drowning and successful recuse case studies.

7.2 Water safety code

The occupier should promote the water safety code, which includes the following:

a) Following safety rules and signs;

b) Never swimming alone;

c) Learning swimming and water survival skills;

NOTE – Refer to Sport Singapore website for information on the SwimSafer Programme.

d) Understanding the dangers of water;

e) Supervising children and weak swimmers at all times; and

f) Swimming in safe areas.

8 Training

8.1 General

8.1.1 Basic information should be provided to all employees (regardless of whether they are full-time or part-time), lifeguards, spotters and volunteers on the location of first-aid kits, availability of trained first aiders and procedures to be followed when first aid is required.

8.1.2 There shall be trained first aiders present at all aquatic facilities.

8.1.3 First-aid training should form part of the regular on-the-job-training.

8.1.4 All employees should be advised and aware of any hazards (including advisories) particular to the aquatic facility.

8.1.5 Hazards shall be isolated or cordoned off wherever possible, with follow-up improvement/rectification works performed to prevent injuries. Any report required should be prepared subsequently in accordance with established procedures.

8.2 Lifeguard

8.2.1 Minimum standard of competency for lifeguards

Lifeguards shall demonstrate a level of competency in which they can consistently apply their skills and knowledge to meet the standards set by the workplace or accrediting organisation. This includes the ability to transfer and apply skills and knowledge to new situations and environments.

NOTE − In Singapore, the Singapore Life Saving Society issues the Pool Lifeguard Award to lifeguards who meet these competency standards.

8.2.2 Certification and recertification

Lifeguards shall maintain their resuscitation and first-aid certifications as current, adhering to the policies of the accrediting organisation. They shall renew the certifications regularly to ensure that they are up to date with the latest techniques and practices in lifesaving and resuscitation.

NOTE − In Singapore, the Singapore Resuscitation First Aid Council is the accrediting organisation for the CPR and AED certifications.

9 Operations manual

9.1 An operations manual should be prepared for every aquatic facility. The manual should be specific to the facility, taking into consideration the design, size, location and activities conducted within the facility.

9.2 All staff should receive a briefing on the manual contents upon commencing employment. They should also receive training pertaining to their key areas of responsibility.

9.3 Each section of the manual should be capable of being issued separately, so that supervisors who are responsible for certain groups of staff can receive the relevant sections.

9.4 The manual is a controlled document and should be reviewed at least once annually or earlier if operational/business needs warrant a review. For the avoidance of doubt, the manual should be dated and version controlled.

9.5 Periodic practice sessions on performing the emergency procedures contained in the manual should be undertaken to ensure all staff understand their roles. At least one evacuation exercise or drill should be conducted annually.

9.6 The operations manual should minimally include details such as the following:

a) Overview of facility layout
   - Facility floor plan
   - Pool dimensions
   - Maximum number of users within the facility
   - Location of alarms, exits, firefighting equipment, first-aid areas/rooms and where appropriate, communication equipment

b) Supervisory procedures
   - Supervision risk assessment
   - Communication
   - Incident control and reporting
   - Emergency response

c) Personnel policies and procedures
   - Lines of responsibility
   - Employee position roles and responsibilities
   - Personnel directory and call tree procedures

d) Training
   - Induction and orientation
   - Qualifications and re-qualifications
   - In-service training

e) Occupational health and safety
   - First aid
   - PPE
   - Incident reporting procedures
   - Hazard identification
   - Isolation and repair
   - Chemical delivery storage and handling
   - Manual handling
   - Safety data sheets (SDS) (see Annex H)
   - Plant inspections and frequency

f) Maintenance
- Plant and equipment
- Buildings
- Fault reporting and repair
- Essential services

g) Water quality
- Operating to relevant regulations
- Turbidity and corrective actions
- Under and overdosing of pool chemicals and respective corrective actions
- Balancing of water quality

h) Programme
- Programme list
- Programme safety requirements
- Pool or room set up and requirements

10 Requirements for oxygen equipment

10.1 The occupier of an aquatic facility that provides oxygen equipment shall ensure that personnel are trained on its use and application in the event of an emergency.

10.2 There should be at least one spare oxygen cylinder stored in an easily accessible location, ready to be used when required.

10.3 The safe storage of oxygen cylinders shall include the following:

a) Secure all oxygen cylinders in racks, stands, or on flat floors to prevent them from tipping over and becoming damaged;

b) Keep oxygen cylinders upright wherever possible. In the event the cylinders need to be stored horizontally, ensure that the cylinders do not roll into other cylinders or objects;

c) Inspect the cylinders regularly for damage or leaks. Contact the supplier/vendor for replacement if necessary;

d) Store oxygen cylinders at least 6 m away from open flames, combustible materials or heat sources;

e) Completely close all valves before storage; and

f) Do not store cylinders in closed spaces such as cupboards, closets, unventilated rooms, etc. Instead, store them in an area that is clean, dry, ventilated, and free from dust, oil, grease, heat and sand.

10.4 All oxygen equipment shall be serviced and maintained per the manufacturer's recommendations or at least annually, or immediately after any problem during operation or when the trained personnel is uncertain of its performance. This information shall be recorded for future reference.

10.5 Oxygen equipment and cylinders, tubing and masks shall be inspected and checked daily for wear and tear, as well as adequacy of oxygen prior to the facility opening. The inspection and checks shall be recorded.

10.6 Staff shall participate in regular practice sessions on the use of oxygen equipment. A record of this training shall be retained.

11 Safety advisory signs

11.1 The occupier shall take reasonable care and responsibility for the safety of all users entering and using their facilities. This can include providing adequate and appropriate signages displayed at prominent areas with heavy footfall, such as near entry points, deep water, sudden drop-off, shallow water and/or within respective pool areas.

The occupier shall display the following pertinent information:

a) Pool rules and regulations;

b) Appropriate behaviours that users shall exhibit in and around pool areas;

c) Importance of pre-participation screening prior to engaging in any activities; and

d) Risks associated with intentional hyperventilation; competitive, repetitive, or prolonged underwater swimming; and breath-holding.

11.2 Other signs that shall be displayed include warnings of slippery floors, cleaning in progress, pool closed, lane closed, and other advisory signage where applicable, particularly in areas with changing conditions such as wave pool, lazy river and feature pool.

11.3 Deep and shallow ends of a pool shall be clearly marked.

11.4 All markings should be of a strong contrast against the surrounding areas. Consideration should be given to developing an overall signage strategy for facilities to minimise the number of individual signs and prevent signage clutter.

11.5 Safety signs should preferably be in pictorial form (see Figure 1). They should be clearly displayed at prominent locations and not obscured, removed or obstructed. Signs should be replaced if they are faded, worn off or damaged in any way that makes them incomprehensible.

11.6 Signage content should be regularly updated to reflect the latest legislation guidelines.

11.7 The plant room shall display the following signages and/or information:

a) Restricted access (e.g. “No entry”, “Authorised personnel only”);

b) No smoking;

c) Hazardous chemical;

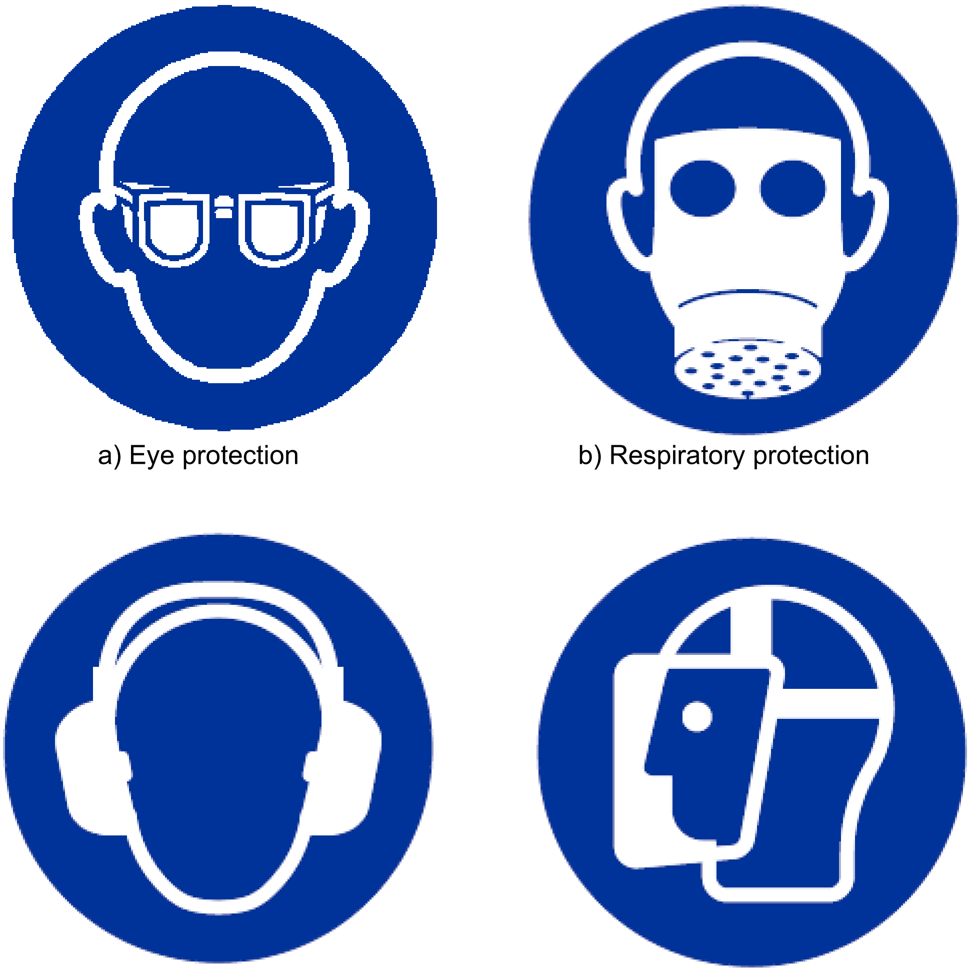
d) Specific chemical labels;

e) SDS;

f) Layout plan of evacuation route;

g) Directional route for evacuation; and

h) PPE requirements.



a) Eye protection

b) Respiratory protection

c) Ear protection

d) Wear a face protection

NOTE – The safety signs are based on the requirements of ISO 7010:2019.

Figure 1 – Example of PPE requirement signages

12 Use of technology

12.1 Technology to aid supervision and security at an aquatic facility can include, but is not limited to: walkie-talkies, emergency alarms, mobile phones, closed circuit television and drowning detection systems. The installation and/or use of technological systems to aid supervision at an aquatic facility should form part of the mitigating measures of the risks identified.

12.2 Technology such as “HELP” activation buttons should be installed in the vicinity of the pool so users or staff can activate it for additional assistance in an emergency. Upon activation, a flashing light should be triggered to inform the lifeguards or spotters of the need for assistance.

12.3 Walkie-talkies provide instant communication between users. Aquatic facilities should consider using walkie-talkies to enhance communication between lifeguards and other facility staff, particularly in circumstances where line of sight amongst staff is not possible.

13 Aquatic facility plant rooms and chemical storage areas

13.1 General

13.1.1 The occupier should allocate space for the storage of all necessary equipment and pool chemicals during the design and planning stage to avoid costly and time-consuming reconfiguration downstream. This planning can facilitate the following:

a) Accessibility for the servicing and maintenance of equipment within the plant room, considering that over time some equipment parts may need to be replaced due to normal wear and tear caused by the necessary use of chemicals to treat the water;

b) Provision of ample space for a clear and safe evacuation route in the event of an emergency; and

c) Prevention of trips and falls.

13.1.2 The occupier should also create space for PPE and installation of emergency eyewash shower stations within the plant room. The PPE should be kept clean and in proper storage shelves or cabinets for easy access.

13.1.3 There should be a separate space requirement for storing equipment that is used for pool operations, such as lane ropes, pool vacuum cleaners, swim pace clock, start blocks, temporary signage (including cones and A-frames), water polo nets, furniture, etc. These items should not be stored in the plant room or chemical storage areas.

13.1.4 To store equipment in a safe and functional manner, the following factors should be considered:

a) Structure of storage space;

b) Personnel who authorised to use the equipment;

c) Weight and size of the equipment;

d) Frequency of use;

e) Position of walkways and emergency exits;

f) Ventilation (e.g. installation of exhaust and intake fan systems); and

g) Trip hazards (failing which the area should be highlighted with warning strips and notices).

13.2 Noise suppression

Equipment in the pool plant room can generate noise. Hence, it is important to consider ways to suppress or block out the noises. The following measures should be considered:

a) Adding a layer of sound insulation around the inside of the plant room to reduce or eliminate background noise;

b) Providing ear plugs for personnel working within the plant room; and/or

c) Monitoring the noise level regularly.

13.3 Refurbishment/modification

13.3.1 The refurbishment/modification of an aquatic facility plant room shall be conducted in accordance with SS 556, including the following:

a) Manual handling and movement of heavy and large equipment or chemicals;

b) Maintenance and removal/replacement of equipment when required;

c) Access and egress during emergencies, delivery and handling of chemicals; and

d) Allowing only trained personnel to access confined spaces.

13.3.2 The plant room shall be furnished with firefighting measures, sheltered, fenced up and under restricted access.

13.3.3 The storage of items within the plant room should not prevent or inhibit access to chemicals, chemical storage areas and/or maintenance of equipment.

13.4 Chemical storage

13.4.1 Hazardous chemicals have the potential to endanger life, thus Annex I on handling of such substances shall apply. There shall be a well-drawn up EAP and proper training to handle the possibility of spillage and accidental release of chemicals to ensure effective containment and minimise dangers to the health and safety of staff, users, public and environment.

13.4.2 A monitoring system should be in place to alert all staff of any leakage of chemicals so that prompt remedial action and evacuation exercises can be activated. Emergency eye-wash stations and safety showers shall be installed in the plant room for any emergencies. Regular checks shall be conducted to ensure that it is functioning properly.

13.4.3 Persons working in the plant room shall be trained in the following areas:

a) Safe handling in the delivery, use and storage of chemicals;

b) Understanding the hazards associated with these chemicals (Annex I and Annex J shall apply);

c) Accessing and understanding the validity of SDS;

d) EAP;

e) Use of appropriate PPE; and

f) Proper housekeeping.

13.4.4 Persons who are working in the plant room shall carry out the following:

a) Prevent chemicals from coming into contact with the water sprayed during cleaning or from roof leaks (see 13.4.5);

b) Store incompatible chemicals in separate elevated areas;

c) Keep chemicals in their original containers, clearly labelled and well protected;

d) Provide barriers or secondary containment walls to hold the total contents of any hazardous chemical, whether in use or in storage;

e) Never mix any pool chemicals;

f) Use a clean, dry bucket, dipper or other container when transferring chemicals;

g) Use the emergency eye-wash and shower station/safety equipment when necessary;

h) Use the buddy system;

i) Observe, understand and internalise instructions such as chemical-feeder use, emergency procedures, cautionary signs, and emergency phone numbers;

j) Ensure appropriate safety and warning signs are installed on all entry doors and spaces; and

k) Discard chemicals following standard operating procedures.

13.4.5 Chemicals shall be contained securely to prevent contact with water, particularly in the event of water leakage into damaged or open containers. Possible sources of water entry include:

a) rainwater from a roof leak or from open or broken windows;

b) wet floor when the stored chemicals were not elevated off the floors;

c) leakage from fire suppression sprinkler system; and

d) hose-down water generated during area cleanup.

13.4.6 Staff shall maintain good housekeeping practices within the chemical storage areas. All combustible and flammable substances shall be kept away from incompatible materials.

14 Supervision

14.1 Children

14.1.1 Regardless of their aquatic ability, infants, toddlers, preschoolers and children are never safe when in or around water and should be under constant adult supervision (see Annex E). They should be accompanied in the water by a parent or caregiver who supervises them and stays within arm’s reach. Parents or caregivers should:

a) keep children aged 0 to 5 years and non-swimmers within arm’s reach;

b) keep children aged 6 to 10 years and weak swimmers close, maintain constant visual contact and be prepared to respond promptly when assistance is needed; and

c) keep children aged 11 to 14 years within visual contact and avoid distractions.

14.1.2 The parent or caregiver is responsible for making sure that the children are in good health while attending an aquatic programme. If there is a known health issue, there should be a doctor’s certificate stating that they are fit to participate in the programme. The parent or caregiver should closely supervise the children and not be involved in other activities, such as reading, using mobile devices, looking away or turning away from the child in the water.

14.1.3 Flotation aids can assist with the gaining of confidence, but they are not life-saving devices and should only be used under adult supervision.

14.2 Special needs groups

14.2.1 There shall be a good ratio of instructor/staff/caregivers to students. Swim instructors shall be trained and qualified. Volunteers and aides shall also be trained for their roles. Spotters can be engaged to look out for any students who wander away from their lessons.

14.2.2 The instructor/staff/caregivers shall assess risks to self and others including equipment used.

14.2.3 Children should be eased into the swim programme, for example, taking them for a tour of the pool before the lesson to give them a chance to get used to the environment.

14.2.4 Instructors/staff/caregivers shall adhere to a safe entry and exit method when conducting lessons in the pool.

14.2.5 The instructor/staff/care givers should consider incorporating suitable equipment into lessons such as kick-boards, preferably with handles, life vests or flotation devices, goggles, water noodles and ear protection.

14.2.6 Audible and visual emergency alarm buttons should be provided within changing rooms.

14.3 Unmanned areas

14.3.1 Unmanned areas, such as changing rooms and toilets, should be checked and inspected regularly for the safety of users as well as prevention of any unsafe acts.

14.3.2 Staff undertaking inspections of these areas should be appropriately trained to respond to emergencies.

14.3.4 Safety hazards identified during inspections should be isolated and sign-posted. Provisions should be made for repairs and/or maintenance to be carried out promptly.

15 Safety requirements for swimming instructors

15.1 General

15.1.1 A swimming instructor’s first priority and responsibility is the safety of their students. They shall take reasonable and practicable measures to reduce risks, and be able to identify and respond to an emergency. They shall hold a valid CPR and AED certificate issued by registered training organisations. Swimming instructors should be registered with the National Registry of Coaches (NROC).

NOTE – Refer to Sport Singapore website for information on the NROC registered coaches.

15.1.2 Swimming instructors should have adequate personal insurance coverage to undertake professional indemnity.

15.1.3 Swimming instructors should provide evidence of their license for teaching swimming especially those teaching specialist groups e.g. infant and pre-school aquatics, and adults or children with disabilities.

15.1.4 Swimming instructors shall closely supervise their students in and around water and when using starting platforms.

15.2 Instructor-student ratio

15.2.1 The information below relates to issues of safety, not ratios considered ideal for teaching effectiveness. The instructor-student ratio applies to learning-to-swim and water safety classes only.

15.2.2 Appropriate instructor-student ratios depend on circumstances such as the following:

a) Facility and its surroundings e.g. pool depth;

b) Student's swimming ability and the presence of physically challenged students;

c) Medical conditions of the student;

d) Type of activity;

e) Skills, ability and experience of the instructor; and

f) Number and skill level of assistants, if present.

The instructor-student ratio should be increased with additional qualified assistants, depending on the evaluation and outcome of the above circumstances.

15.2.3 In addition to the certified swimming instructor undertaking the swimming and water safety instruction and who is responsible for the students, a second person capable of providing assistance in the case of an emergency should be available.

15.2.4 For the safety of students, the following maximum ratios are recommended for learning-to-swim and water safety lessons conducted in swimming pools:

a) When teaching beginners with little or no experience in shallow water, the maximum instructor-student ratio for a swimming pool is 1:10.

b) When teaching intermediate students who have basic skills and can swim 25 m with a recognisable stroke and can demonstrate comfort and confidence in the aquatic environment, the maximum instructor-student ratio for a swimming pool is 1:12.

c) When teaching advanced students who can swim 50 m using two recognisable strokes, demonstrate one survival stroke in deep water and display comfort and confidence in the aquatic environment, the maximum instructor-student ratio for a swimming pool is 1:15.

d) The maximum instructor-student ratio for infants/toddlers (6 – 36 months) is 1:6 (accompanied by the infant's parent in water).

e) The maximum instructor-student ratio for preschoolers (36 – 42 months) is 1:4.

f) The maximum instructor-student ratio for preschoolers (42 – 60 months) is 1:5.

15.2.5 Special groups should require specific attention. A needs analysis should be conducted. The instructor-student ratio in those cases should be 1:1, depending on the circumstances.

15.3 Certified swimming and water safety instructor

15.3.1 Each certified swimming and water safety instructor shall be positioned in a way that allows them to maintain a constant view of all students and respond promptly when assistance is needed.

15.3.2 The instructor shall ensure that all their students are safely out of the swimming pool at the end of the lesson. An adult volunteer, e.g. a parent, should be engaged to look after students whose parents are late in picking them up after the lesson.

15.3.3 The instructor should be familiar with the locations and placements of rescue and lifesaving equipment such as rescue tubes, ropes, poles, approved buoyancy devices and AED that are used for emergencies.

15.3.4 The instructor should adopt and implement the relevant practices for child protection as stipulated by Safe Sport.

NOTE − Refer to Sport Singapore website for information on the Safe Sport.

16 Hire of facilities

16.1 The facility hirer (or part thereof) should receive from the management a letter of agreement or contract that clearly establishes the responsibilities of both parties. This contract should be countersigned by the hirer to indicate that its terms and conditions have been accepted.

16.2 The contract should clearly establish the following:

a) Information regarding the specific number of persons using the facility and their swimming skills;

b) Activities to be conducted by the hirer should also have a letter of agreement if the activities are conducted by an external party;

c) Duration of hire;

d) Name of the hirer, and/or the hirer’s representative who can be personally present and in charge of the group;

e) Age of the hirer and hirer’s representative;

f) Party (whether the aquatic facility or the hirer) providing lifeguards to supervise the session;

g) Number of lifeguards to be present and assigned to the group during the session (and qualifications of the lifeguards, if provided by the hirer);

h) Party (whether the aquatic facility or the hirer) providing first aid for the session;

i) Number of first-aid providers to be present during the session (and qualifications if provided by the hirer), which should satisfy 8.1.2;

j) Respective responsibilities of the facility management and the hirer in an emergency. A distinction should be drawn between general emergencies and facility emergencies (e.g. structural issues/integrity);

k) Party (whether the aquatic facility or the hirer) responsible for insuring the activity;

l) Enforceability of any local laws;

m) Rules of behaviour and conduct to be followed;

n) Specific advice to be given to participants;

o) Location and access to first aid and/or rescue equipment; and

p) Relevant medical conditions of participants e.g. allergies.

16.3 The management of the aquatic facility is responsible for ensuring that the hirer’s stated activities in the contract is compliant with the facility’s risk management plan.

16.4 The hirer should be provided with a copy of the EAP and any related procedures. Hirers should be requested to sign and acknowledge that they have read and understood the information. Details of documentation, training provided and checks should be recorded.

16.5 Hirers who require a risk assessment prior to conducting their activities at an aquatic facility should conduct their own risk assessment and focus on reducing risks at the site of the activities. Hirers should also take reasonably practical measures to protect the participants using the facility.

17 Inclement weather

17.1 It is unsafe to remain outdoors and/or swim in any aquatic facility whenever there is lightning activity within the vicinity.

17.2 Outdoor aquatic facilities should be temporarily closed when there is lightning activity. Users of outdoor swimming facilities should be evacuated to a covered area that has sufficient electrical earthing in case of a lightning strike. Children should be closely monitored and supervised.

17.3 There should not be any use of electrical equipment during thunderstorms.

NOTE – Refer to the Meteorological Service Singapore website for weather advisories.

18 Interactive water play equipment

18.1 Interactive water play equipment is similar to those found in children's playgrounds, but the water environment introduces additional risks for users. Given the enthusiastic nature of children, increased supervision is essential.

18.2 When the facility is open for use, the pool should be constantly supervised and equipment checked regularly.

18.3 Before using the equipment, a daily inspection should be conducted to detect any damages that can pose a safety risk to users. If any hazards are identified, they should be immediately isolated and reported for repair using standard operating procedures.

18.4 All structural recommendations, for example, stating the maximum number or height of people on the equipment at any one time, should be adhered to.

18.5 A schedule of rules governing the use of the equipment should be established and monitored. The rules should include, but are not limited to, the following:

a) Maximum number of people;

b) Permitted age ranges and height;

c) Disallowing climbing or running on the equipment;

d) Necessary adult supervision:

e) Swimming capabilities of people using the equipment; and

f) Proper use of equipment.

18.6 Barricading of any components or equipment can be needed for several reasons, including safety, maintenance and crowd control. The isolation of any component or equipment should be carried out effectively, ensuring safety, and appropriately signed for by the management.

18.7 Water pressure changes resulting in high pressure jets of water or water spraying out should be monitored and any necessary corrective action should be initiated immediately.

18.8 Appropriate safety advisory signage should be displayed on the use of the equipment, and any possible dangers involved. The signage should also satisfy Clause 11, where appropriate.

18.9 Users of water play features, e.g. swinging apparatus, should adhere to the safety rules stated in 18.5, such as on the appropriate height, weight and the number of users.

18.10 An EAP relevant to the play equipment should be in place. There should also be audible and visual emergency alarm buttons located near the equipment.

19 Use of inflatable play equipment

19.1 It is presupposed that the use of inflatable play equipment are in compliance with applicable statutory and regulatory requirements. In the absence of this, the manufacturer's guidelines should be adhered to. The occupier should also conduct a risk assessment on the use of such inflatable equipment in the aquatic facilities.

19.2 The risk assessment should include, but are not limited to, the following:

a) Manufacturer's instructions;

b) Depth of water in which the inflatable is operated;

c) Likelihood of a person falling off or onto the inflatable, then into the water, or onto the pool edges;

d) Design of the pool;

e) Design of the inflatable;

f) Physical size of users;

g) Number of individuals using inflatables within the facility;

h) Physical ability of users;

i) Operating procedures;

j) Supervision of the inflatable and the general area of use (a constant line of sight between supervisors and users should be maintained at all times);

k) Crowd management; and

l) Previous incidents in relation to the inflatable, either recorded or anecdotal.

19.3 If the occupier assesses an unacceptable risk, the use of the relevant inflatable should cease immediately until an acceptable means of managing this activity is implemented.

20 Electrical safety

20.1 Work on electrical installations and equipment requires specialised skills and shall only be performed by approved or registered employees or tradespeople.

20.2 Electrical equipment shall be kept away from water and aquatic facilities.

20.3 In situations where it is necessary to use electrical equipment near aquatic facilities, the following precautions should be taken:

a) The electrical socket outlet into which electrical equipment is to be connected should be protected e.g. by using a safety switch;

b) The electrical socket outlet should be installed at least 3 m away from the nearest pool and at least 1 m above the ground level;

c) Electrical equipment should not be left unattended;

d) Pneumatic-operated equipment should be used, whenever possible;

e) Electrical cords and equipment should be highlighted and preferably kept clear of concourses to remove any chance of electrocution or injury as a result of a trip or fall; and

f) The aquatic facilities should be vacated when electrical equipment is to be used nearby.

20.4 Workers should not use or repair any electrical cord, fixture, terminal box or equipment while standing in water or on a wet concourse.

20.5 Aquatic facilities should have their electrical equipment ''tested and tagged'' by an appropriately qualified person on a regular basis.

20.6 For proper planning, allocation and placement of power sockets, the facility owner/occupier shall determine:

a) the number and placement of power sockets required for the facility, instead of overloading a single power extension socket to prevent fire risk; and

b) the proper placement of sockets within the facility, in particular, the plant room and near water dispensers.

Annex A
(informative)

Major and minor incident templates

A.1 The following shows a sample of a major incident report.

PRELIMINARY – MAJOR ACCIDENT REPORT
(To be submitted within 12 hours of incident)

Definition of Major / Minor Accident, Rescue & Ambulance Activation

Event	Definition
Major Accident	Death of any person, serious head/facial/body injuries, deep cuts with continuous bleeding, skeletal injuries (serious bone fractures & dislocations, spinal injury), amputation, eye injuries resulting in loss of sight (permanent or temporary), and/or any injury resulting in unconsciousness, resuscitation or admittance to the hospital
Minor Accident	Irritation, ill-health with temporary discomfort, abrasion, bruises and minor cuts requiring first aid treatment only.
Rescue	Rescue activities carried out at that instant to prevent/minimize/isolate the victim from the danger zone/activities to avoid further injury. (Classification of a rescue under Major or Minor Accident requires the assessment of the severity of the victim's condition)
Ambulance Activation	Classification of an ambulance activation under Major or Minor Accident requires the assessment of the severity of the victim's condition, follow-up treatments and/or hospitalization for observation.

Note: Sports Centres' Management to consult their Supervisor when in doubt.

☑ *Tick where applicable*
☐ Land Accident ☐ Water Rescue   Day/Date: ____________

Sports Centre: ______________________________   Weather: __________

Victim's Particulars:

Name: ______________________________   Sex: ☐ M ☐ F   Age: ______
Address: ______________________________   Postal Code: ____________
NRIC No.: ________________   Nationality: ☐ Local ☐ Foreigner: ____________
Contact No.: ________________   ☐ Res ☐ Off. ☐ HP   Profession: ____________
Race: ☐ Chinese ☐ Malay ☐ Indian ☐ Eurasian ☐ Others: ______________
Name/Contact No. of Next-of-Kin/Friends:
____

Details of Incident:

☐ Weak Swimmer ☐ Non-Swimmer ☐ Others: ____________
Location: ____________   Time: ____________   No. of Swimmers: ____________

Condition of victim: ☐ Conscious ☐ Unconscious ☐ Agonal Breathing
Rescued by: ☐ Staff ☐ Others   Name/Contact No. of Rescuer: ____________
CCTV Recording: ☐ No ☐ Yes

______________________   ______________________
Name/Signature: ☐ Injured ☐ Rescued Person   Name/Signature of Reporting Officer

Detailed Major Accident Report To Be Submitted Within 3 Working Days

Treatment Efforts By Staff:

Do not remove AED battery

AED shocks: ☐ No ☐ Yes No of shocks: __

CPR efforts: ☐ No ☐ Yes

Oxygen Admin: ☐ No ☐ Yes

Ambulance Activation:

Ambulance Activated: ☐ No ☐ Yes

Ambulance No: ________________

Treatment Efforts By Paramedics:

AED shocks: ☐ No ☐ Yes No of shocks:__

CPR efforts: ☐ No ☐ Yes

IV: ☐ No ☐ Yes

Oxygen Admin: ☐ No ☐ Yes

Any Other Treatment:

____

Hospital :____________________

Police Activation:

Police Activated:

☐ No ☐ Yes IO's name: ____________________ Contact No: ____________

Witness Information:

Name : __________________ Contact No:_________________

Name : __________________ Contact No:_________________

Centre Manager / Asst Centre Manager's Report

____

____

____

____

____

Designation : ________________ Signature : ________________ Date : ________________

Director / DD / AD Report

____

____

Verified & closed by: ________________ ______________ ____________

Name / Designation Signature Date

Other Information:

Victim's Height: __________ Build: __________ Abnormality: __________

Swimwear Color: ______ ☐ One pc ☐ Two pc ☐ Others ________________

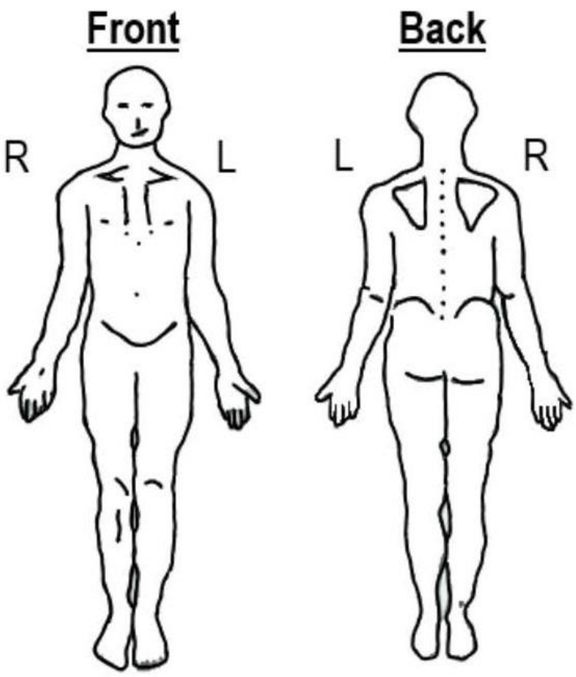
Dentures: ☐ No ☐ Yes

Personal belongings: ☐ No ☐ Yes

Locker key on wrist: ☐ No ☐ Yes

Victim's Medical History:
(if applicable) ________________________________

Please mark a cross(es) (X) at injured area on diagram below



A.2 The following shows a sample of both major and minor incidents in a report.

PRELIMINARY – MAJOR/MINOR* INCIDENT REPORT

*Delete where appropriate

(For major incidents, the officer to follow-up with Detailed Major Incident Report)

Definition of Major/Minor Incident, Rescue & Ambulance Call

Event	Definition
Major Incident	Fatal, head and face/facial injury, serious body injury, deep cuts with continuous bleeding, fractures and suspected spinal injury, total permanent disability and in a state of unconsciousness.
Minor Incident	Irritation, ill-health, with temporary discomfort, bruises and minor cuts requiring only first-aid treatment.
Rescue	Rescue activities carried out at that instant to prevent/minimise/isolate the victim from the danger zone/activities to avoid further injury. (Assessment on the severity of the victim's condition to classify the rescue leading to a Major or Minor incident)
Ambulance Call	Classification of Ambulance Call as Major or Minor accident depends on the severity of the victim's condition for follow-up treatments under the definition of Major or Minor incident

Note: Officer to seek advice from their supervisor for classifications when in doubt or handling complicated accident.

Tick where applicable ☐ On Land ☐ In Water Reference No: ______

Facility: ______ Date: Day Mth Yr

Victim's Particulars:

Name: ______ Sex: ☐ M ☐ F Age: ______

Address: ______ Singapore ______

NRIC No: ______ Nationality: ☐ Local ☐ Foreigner

Contact No: ______ ☐ Res ☐ Off. ☐ Hp ☐ Pg Profession: ______

Race: ☐ Chinese ☐ Malay ☐ Indian ☐ Eurasian ☐ Others: ______

Details of Incident: Brief description of what happen ______

Location: ______ Time: ______ No. of Users/Patrons: ______

Condition of victim before attend: ☐ Conscious ☐ Semi-conscious ☐ Unconscious

Rescued by: ☐ Staff ☐ Public Name and contact no. (of rescuer) ______

Signature of ☐ Injured ☐ Rescued Person Name and signature of reporting officer

Ambulance Summoned ☐ No ☐ Yes: QX: ______ Hospital: ______

Police Notified: ☐ No ☐ Yes: (If yes), Name of officer: ______ Car No. QX ______

Witness: ______________________ Address/Tel: ______________________

Senior Centre Manager/Centre Manager/Assistant Centre Manager's (SCM / CM / ACM's) report

____

____

____

____

____

____

Designation: __________________ Signature: __________________ Date: ______________

Assistant Director / General Manager / Assistant General Manager's (AD / GM / AGM's) report

____

____

Verified & closed by: __________________ __________________ ______________

Name/Designation Signature Date

Other information

Victim's Height: ______________ Build: Big/Medium/Small Abnormality: ______________

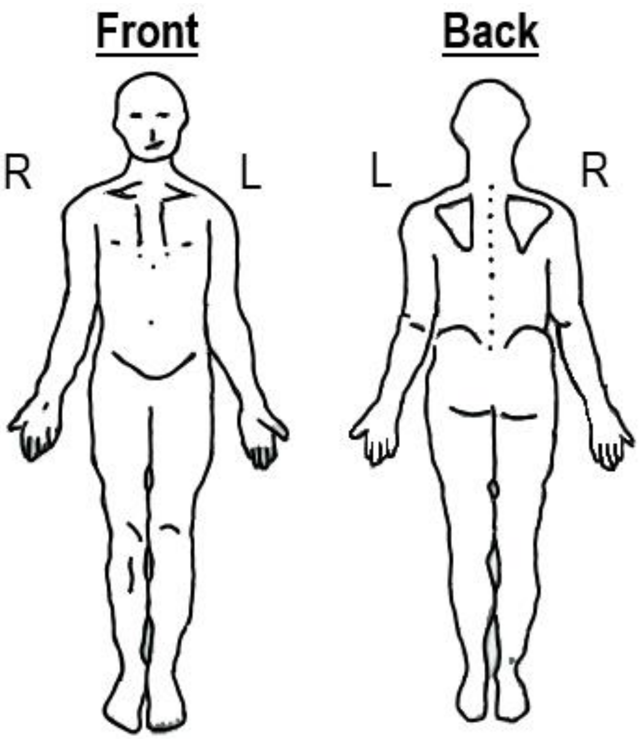
Swimming Suit / Dress Colour: ______________ ☐ One-piece ☐ Two-piece ☐ Others

Victim's Medical History:
(if applicable) ____

____

____

Please mark a cross(es) (X) at injured area on diagram below



Annex B
(informative)

Example of emergency action plan (EAP)

1 Introduction

The purpose of these procedures is to outline the actions to be taken in the event of an emergency in the pool and around the poolside areas.

The facility manager shall ensure all staff are aware of the responsibilities with respect to these procedures.

2 Responsibilities

Responsibility for carrying out emergency actions rest with the lifeguards and other pool staff.

The lifeguards are responsible for controlling the incident/accident and for making the decision to evacuate the pool.

3 Procedure

3.1 Raising alarms

The method of communication using a whistle is as follows:

a) One whistle blow – to attract the attention of pool users;

b) Two whistle blows – to attract the attention of other pool staff;

c) Three whistle blows – to indicate that the lifeguard is about to take emergency action; and

d) One long whistle blow – to attract the attention of pool users to prepare for an evacuation.

Whistles will be used sparingly. Relevant verbal instructions, hand signals or walkie-talkies can be used as well.

3.2 Minor emergencies

Minor incidents or emergencies, if handled properly, will not result in a life-threatening situation. Examples of incidents of this nature include a pool user slipping at the poolside, a minor cut or bruise and a reaching rescue. Whilst these incidents can seem routine, they can pose an increased risk of a more serious incident if proper procedures are not followed. To ensure an appropriate response, the lifeguard, on becoming aware of the incident should follow this procedure:

a) Notify other pool staff that they have to respond to an incident;

b) Instruct other pool staff move to a covered area or request additional assistance if necessary;

c) Look for a first aider to administer aid or provide appropriate assistance;

d) Refer any casualty to an appropriate location; and

e) Complete an accident/incident report as necessary.

3.3 Major emergencies

A major emergency is defined as an incident resulting in a serious injury or a life-threatening situation. Typically, multiple staff members are involved, and in extreme situations, the entire team is required to provide support. The procedure for dealing with major emergencies is as follows:

a) The lifeguard will raise the alarm by using a whistle blow alarm (see 3.1), blowing three whistle blows and/or using hand signals;

b) If the office has not been notified, the nearest member of staff will notify management of this emergency;

c) The lifeguard will initiate rescue/first aid and remove any casualty from the area;

d) The support team members will cover the area vacated, assist the lifeguard and evacuate the pool if necessary;

e) The lifeguard will ensure an ambulance is requested, supplies specialist equipment and takes control of the situation, including managing and assisting other users;

f) A member of staff will be assigned to meet the ambulance crew to brief them and escort them to the scene of the incident;

g) The ambulance crew is assigned the responsibility once they start to treat the casualty;

h) The lifeguard will ensure that safe levels of supervision are maintained for the duration of the incident and subsequent action; and

i) The lifeguard will ensure that all accident/incident reports are completed and the necessary follow-up action is taken.

Actions to be taken in the event of specific emergencies are detailed in 3.4 to 3.7.

3.4 Fire alarm

The procedure for raising the alarm is follows:

a) Break glass units are located in the sports facility but the sports office can be contacted via walkie-talkies to initiate an alarm; and

b) On hearing the alarm, the member of staff in charge of the party shall initiate evacuation to the designated safe fire holding area.

Once the alarm has been raised, those at the poolside should blow their whistles as per the prevalent centre standard operating procedures and clear the pool as quickly as possible.

Everyone should be directed to the nearest emergency exit. Thermal blankets will be issued if necessary and will be brought to the assembly point by the nominated first aider at the poolside (see 3.13).

3.5 Discovery of a casualty in the water

The first response to a casualty in the water should be to consider performing a rescue by reaching with a pole or rope. Whenever possible, hand-to-hand contact should be avoided until the casualty is under control and the possibility of another person being pulled into the water is reduced.
The pool should only be evacuated if necessary.

The lifeguard should only enter the water to affect a rescue if other alternatives are not feasible.

If entry into the pool is necessary, the procedure to be applied is as follows:

a) To attract the attention of another lifeguard or additional support by using the pool alarm and/or blowing the whistle loudly three times;

b) If the poolside drown alarm has not been activated or alarm raised, the nearest member of staff to the alarm should activate/initiate it;

c) If the lifeguard is carrying a radio, it should be placed at the poolside prior to entry;

d) The lifeguard should enter the water in a safe manner, recover the casualty and land them at the nearest suitable landing point; and

e) The lifeguards will follow resuscitation protocols in accordance with National Pool Lifeguards Qualifications (NPLQ) and/or first-aid training until the ambulance crew takes over.

3.6 Serious injury to a user

General

The process for dealing with major emergencies as detailed in 3.3 will be followed in the event that the pool staff notices a user with a serious injury. The lifeguards will follow first aid/resuscitation protocols in accordance with NPLQ or first-aid training. These will be followed until the ambulance crew takes over. In cases of serious injury, unconsciousness or suspected broken bones, patients will not be moved until first aid has been given by the ambulance crew.

Head injuries

All head injuries will be treated as serious injuries and lifeguards will follow first aid/resuscitation protocols in accordance with their NPLQ or first-aid training. In addition to the major emergency procedures outlined in 3.3, casualties with face/head injuries should not be allowed to return to the pool.

3.7 Dealing with blood, vomit and faeces

In the event that blood, vomit and faeces are discovered in the pool or at the poolside, the following procedures will be applied:

Blood

a) If substantial amounts of blood are spilled into the pool, it will be temporarily cleared of people to allow the pollution to disperse and any infectious particles within it to be neutralised by the disinfectant in the water;

b) When clearing blood, the correct personal protective equipment, i.e. disposable gloves, shall be worn; and

c) Spillages of blood at the poolside will be contained, and covered in paper towels to enable the towels to soak up the blood and wiped up immediately. Blood will not be washed into the pool or poolside drains. Soiled towels will be disposed of properly in clinical waste bins, for example, nappy bins. The area will then be disinfected.

Vomit

a) If substantial amounts of vomit are spilled into the pool the affected pool will be closed to users in order to allow for its removal;

b) The vomit will be removed from the water using a scoop and placed in a bucket, the contents of which will be flushed down the toilet;

c) A minimum of three turnover periods of the affected pool will elapse to ensure the removal of any bacteria;

d) Prior to the pool re-opening, conduct a water quality test to ensure that chlorine levels and total dissolved solids (TDS) levels are within the agreed parameters, along with a visual inspection;

e) When clearing vomit, the correct personal protective equipment, i.e. disposable gloves, shall be worn;

f) Spillages of vomit at the poolside will be contained, covered in paper towels to enable the towels to soak up the vomit as much as possible and wiped up immediately. Vomit will not be washed into the pool or poolside drains. Soiled towels will be disposed of properly in clinical waste bins, for example, nappy bins. The area will then be disinfected; and

g) Any equipment that has been used to scoop up the vomit shall be thoroughly disinfected before it is stored away.

<u>Faeces</u>

If faeces is discovered in the pool, the affected pool will be closed immediately to allow for its removal.

3.8 Human waste

If there is evidence of human waste in the water, the pool shall be vacated immediately. The pool will then be treated fully, as described in 3.7.

3.9 Epileptic fit

Should a user suffer an epileptic fit whilst in the pool, a staff member should enter the water and support the user, holding their face clear of the water until the fit ends. The staff member should not attempt to remove the user from the water. Once the fit has finished, the user should be removed from the pool and be given the appropriate first aid.

If the epileptic fit persists for more than 2 min, the staff member should seek additional assistance to guide the user towards the pool edge to be removed from the pool. Emergency medical services should be contacted immediately in such instances.

The remainder of the users should be made to evacuate the pool and enter the changing rooms as soon as it is possible.

3.10 Asthma attack

If a user suffers an asthma attack, they should be removed from the pool, be administered their inhaler and then treated as normal. Help should be summoned from the sports office.

3.11 Emergency procedures / notices

All emergency procedures and relevant notices should be brought to the attention of users when they first visit the pool and highlighted on subsequent visits.

Emergency procedures to clear the pool should be practised during the first swimming session and regularly thereafter.

3.12 Handling potential emergencies

In the event of a user suffering an accident and potentially becoming unconscious in the water, these procedures should be followed:

a) The staff should move all other pool users to the side of or out of the pool;

b) Sound one long loud whistle blow as the emergency signal to clear the pool;

c) Staff should facilitate a rescue, using necessary equipment and methods;

d) After safely removing the user from the pool, administer appropriate first aid, including CPR, and call for assistance;

e) If an ambulance is required, call 995 using a mobile phone, or if the signal fails, contact the secretary or caretaker via walkie-talkie to call an ambulance using the landline;

f) Staff can also use the air horn to attract help;

g) As soon as possible, relocate the remaining users to the changing rooms; and

h) The injured user should be removed from the swimming pool area as needed, depending on their injuries (see 3.5).

3.13 Fire evacuation

If the fire alarm sounds, staff members should follow these steps:

a) Evacuate the pool;

b) Gather the users and conduct a roll call;

c) Instruct users to put on their shoes and towels and to collect their belongings; and

d) Contact the sports office via walkie-talkie to identify the fire’s location, determine the designated muster point and confirm that all class members are all accounted for.

3.14 Storm

If a storm should gather, the users should be evacuated from the pool, in case of a lightning strike.

3.15 Reporting incidents

Any accident, near misses, or health and safety issues shall be immediately reported to centre management staff. There is an incident record book in the office, which should be completed whenever such incidents occur, and the details shared with centre management.

4 Other foreseeable emergencies

Take appropriate action in the event of a foreseeable emergency, for example:

a) overcrowding;

b) disorderly behaviour (including violence to staff);

c) water clarity issues;

d) fire outbreak (or sounding of the alarm to evacuate the building);

e) bomb threats;

f) lighting failures;

g) structural failures;

h) emission of toxic gases;

i) serious injury to a user; and

j) discovery of a casualty in the water.

5 Effectiveness of emergency procedure

The procedure should be clear, especially the procedure for evacuation from the pool or building. To ensure the effectiveness of emergency procedures, management should:

a) adequately train all staff on such procedures;

b) display notices to advise the users of the arrangements;

c) regularly check the exit doors, signs, fire-fighting equipment and break-glass call points where provided, to ensure they are free from obstruction; and

d) ensure all fire exit doors are always operable without the aid of a key when the premises are occupied.

6 Hire of facilities

When hiring the pool to outside organisations, ensure that the contract includes the following:

a) Information on the number of participants and their swimming skills;

b) Name of hirer’s representatives in charge of the group;

c) Numbers and skills/qualifications of certified safety professionals to be present during the session; and whether they will be provided by the hirer or by the pool operator;

d) Copies of normal and emergency operating procedures, with a signed acknowledgement of understanding to be provided to the hirer;

e) An agreement on the responsibilities of the pool operator and the hirer during emergencies. A distinction needs to be made between emergencies arising from the activities of the group using the pool and other emergencies (e.g. structural or power failures, etc.);

f) Requirement that the pool operator maintains responsibility for non-group-related emergencies, and provides competent staff during hire sessions;

g) Any rules of behaviour to be enforced during the session; and

h) Any safety advice to be communicated to participants, such as avoiding alcohol and food immediately before swimming.

Annex C
(informative)

Risk assessment

C.1 Risk assessment template

C.1.1 An example of a risk assessment template is given in Figure C.1.

Activity Description:	**Risk assessment ownership:**		**Initials and Org Stamp:**	**Issue**	**Approved by:**	**Issue date: (DD/MM/YYYY)**	**Initials**
Poolside Party	Activity OIC:	Mrs ABC					
	Organisation:			1			
Location:	**Risk assessment team members**						
ABC Condo Pool	RA Leader:	**Mrs ABC**		**Issue**	**Vetted by:**	**Issue date: (DD/MM/YYYY)**	**Initials**
	RA Members:	Mrs DEF					
Group description	RA Members:	Mrs GHI		1			
25 children below 7 yrs old and 10 adults	RA Members:						
	RA Members:			**Date of next review:**			

Type of Activity													
Hard Identification			**Inherent Risk**						**Control measures and residual risks**				
Step	**Description of Activity**	**Possible Hazard**	**Potential Injury (Effect)**	**Severity**	**Cause of Effect**	**Existing Control Measures**	**Likelihood**	**Risk**	**Additional Control Measures**	**Action Owner**	**Severity**	**Likelihood**	**Residual Risk**
1	Swimming	Water	Drowning	4	Death or comatose	Ensure that children are wearing jackets	2	8	Appoint a parent to 3 kids to closely supervise them at arms length	Mrs ABC	2	1	2

Figure C.1 – Risk assessment template

C.2 Categorisation of risk assessment

C.2.1 Water safety

Assessment of layout including general and surrounding areas, shower facilities, pool deck, organisation and equipment that can pose safety risks to users.

C.2.2 Workplace safety

Evaluation of the layout, organisation and serviceability of equipment that can impede safety and potentially harm users.

C.2.3 Fire safety

Analysis of the placement, layout and serviceability of fire equipment, including evacuation routes, that can cause delays, interruptions or harm to users during emergencies.

C.2.4 Occupational health and safety

Prolonged exposure or interaction with situations that can lead to strain or side effects for users.

C.3 Severity and consequences

Severity and consequences are the degree or extent of injury or harm caused by accidents or incidents arising from a hazard. It can be classified into five categories, as shown in Table C.1.

Table C.1 – Severity and consequences

Level	Severity/Consequences	Description
1	Insignificant	• No injury, only minor discomfort • Insignificant environmental damage
2	Minor	• Minor injury such as cuts and bruises • Localised environmental damage • First-aid treatment
3	Moderate	• Medical treatment required • Environment damage affecting organisation
4	Major	• Single fatality • Permanent major disability • Environmental damage affecting entire site/location
5	Catastrophic	• Multiple fatalities and severe injuries • Widespread environmental damage affecting off-sites

C.4 Likelihood

Likelihood describes the occurrence of an accident/incident or ill health. It can be classified into five categories, as shown in Table C.2.

Table C.2 – Categories of likelihood

Level	Likelihood	Description
1	Rare	Unlikely to occur, very low probability
2	Unlikely	Can occur at some time
3	Moderate	Occurs occasionally, but unexpectedly
4	Likely	Occurs occasionally and to be expected
5	Definitely	Common or repeated occurrence

C.5 Risk matrix

C.5.1 To minimise the subjectivity when determining severity and likelihood, the following sources of information should be considered:

a) Existing controls;

b) Past incidents and accident records;

c) Industry practice and experience; and

d) Relevant published literature.

C.5.2 Once the severity and likelihood have been established, the risk level is determined by the 5x5 risk matrix, as shown in Table C.3.

Table C.3 – 5x5 risk matrix

Likelihood \ Severity	1 Insignificant	2 Minor	3 Moderate	4 Major	5 Catastrophe
1 Rare	1	2	3	4	5
2 Unlikely	2	4	6	8	10
3 Moderate	3	6	9	12	15
4 Likely	4	8	12	16	20
5 Definitely	5	10	15	20	25

C.6 Risk control table

After determining the risk level, risk controls should then be in place to reduce the risk level to an acceptable level. This can be done by using the risk levels and control actions table (see Table C.4) to guide the selection of risk controls to reduce the severity, consequences and/or likelihood.

Table C.4 – Risk levels and control actions

Risk level	Risk control actions and timescale
1-3 (Low risk)	• No action is required • No monitoring is necessary
4-5 (Low risk)	• No additional controls are required • Monitoring is required to ensure existing controls are effectively implemented
6-9 (Medium risk)	• Efforts should be made to reduce the risk • Risk reduction measures should be implemented within a defined timeframe • Management control is required
10-14 (Significant risk)	• Work should not commence until the risk has been reduced • Considerable resources should be allocated to reduce the risk • Where work has commenced, urgent interim measures shall be taken • Senior management action is required
15-25 (High risk)	• Work shall not start or work shall be suspended immediately • If it is impossible to reduce the risk, work shall remain prohibited • Detailed research and management plan are required

C.7 Risk control

It is essential to eliminate or reduce risks at their source. If a risk cannot be controlled completely by engineering measures, protect the employees through administrative control or personal protection. Employing the hierarchy of controls enables effective hazard control and risk reduction (see Table C.5).

Table C.5 − Hierarchy of controls

Hierarchy		**Description**	**Examples**
	Elimination	Total removal of hazards	– Remove sharp edges or corners – Eliminate manual handling
	Substitution	Replace the hazard by one that presents a lower risk	– Substitute a toxic substance with a milder one – Divide a load to make it easier to handle
Most Effective	Engineering controls	Physical means to limit hazards	– Consist of enclosure, isolation, limitation and containment
Least Effective	Administrative controls	Adhere to procedures or instructions	– Consist of procedures or work practices and subject to users' compliance
	PPE	Use of PPE only as a last resort as it aims to protect personnel when all other control measures are not practical and subjected to users' compliance	– Use protective gloves when handling chemicals

Annex D
(informative)

Types of hazards

Types of hazards	Examples
Mechanical – Moving parts of a machine with rotary, sliding or reciprocating motion of a machine can have sufficient force in motion to cause injury (such as crushing, fractures, amputation) to people.	– Entanglement resulting from: i. contact with a single rotating surface such as coupling, spindles, bars or any rotating work piece. ii. being caught by projection in gaps such as fan blades, belt or pulley, chain wheels and flywheels. iii. hands caught in between counter-rotating parts such as gear wheels, rolling mills. – Cutting, shearing and crushing actions of a machine. – Draw in by in-running rip points between two counter-rotating parts.
Electrical – Any contact with electricity is dangerous, especially when the electrical current passes through the body.	– Damaged or faulty electrical outlet or plug. – Exposed live wires from electrical wiring or connection. – Overloading of electrical supply, wiring or system. – Contact with electrical parts with wet hands or laying electrical wire on wet ground.
Chemical – Certain chemicals cause immediate injury such as chemical burns, acute poisoning or long-term effects such as cancer with eventual death. Also, they can cause hazards such as fire and explosion.	– Contact with corrosive chemical causing burns. – Entry of chemicals into body through inhalation, ingestion or absorption causing acute poisoning or long-term health effects. – Fire and explosion from flammable chemicals. – Death due to exposure to extremely toxic chemicals.
Slip, trip and fall – Poor ground conditions, wrong selection of footwear and poor housekeeping contribute to majority of the slip, trip and fall hazards.	– Slippery floor or poor ground conditions. – Poor housekeeping causing obstruction and slipping/tripping hazards.
Fire – Fire outbreak due to accumulation of flammable/combustible materials and poor heat source control. – Accidental release of huge amount of explosive mixture, causing an explosion.	– Hazardous accumulation of combustible/flammable materials. – Poor and uncontrolled storage of flammable chemicals. – Overloading of electrical appliances or system.

Types of hazards	Examples
Deep water	Weak swimmers who are not proficient can face issues such as drowning.
Sudden drops	Both deep end and shallow end share similar base colour, thus swimmers are unable to identify the sudden change in depth.
Lack of safety and rescue equipment	Pools without safety and rescue equipment are extremely unsafe.
Poor water quality	Poor water quality in swimming pool can lead to health problems.
Lack of supervision	The lack of supervision is dangerous, even for a proficient swimmer as no one can be there to lend a helping hand.
Disease	Swimmers with open wound can spread diseases to others.
Distractions	Anyone who has been tasked to look after another individual when in swimming pool but failed at exercising this responsibility, e.g. playing/looking at electronic devices, reading, talking, etc.

Annex E
(informative)

Tips to prevent drowning

Some tips to prevent drowning are given below:

a) Learn swimming and survival skills, such as the SwimSafer programme.

b) Always supervise children within arm's length while in the pool. Avoid distractions like reading or using cell phones.

c) Be aware of the pool depth and obey "no diving" restrictions in shallow areas.

d) Familiarise yourself with lifesaving equipment like life buoys and rescue tubes.

e) Observe safety rules and messages, avoid horseplay and practise good pool etiquette.

f) Swim with family members or buddies to watch out for each other.

g) Install pool fences or barriers.

Annex F
(informative)

Slip, trip and fall hazards and prevention

F.1 Running on wet surfaces within the swimming pool can result in slips, trips and falls. Injuries such as concussions, serious cuts on the head, and fractures can occur.

F.2 Some slip, trip and fall hazards and its prevention are as follows:

a) Walk slowly instead of running on wet surfaces;

b) Install anti-slip or warning strips on slippery areas;

c) Safely secure any wiring to prevent tripping; and

d) Minimise wet areas around the swimming pool.

Annex G
(informative)

Swimmers' behaviours

The following are some swimmers' behaviours:

a) Active drowning where swimmer is struggling to keep the mouth above the water surface in order to breathe (see Figure G.1);

b) Passive or silent drowning (see Figure G.2);

c) In distress, where swimmer is yelling or waving arms (see Figure G.3).

Figure G.1 – Active drowning

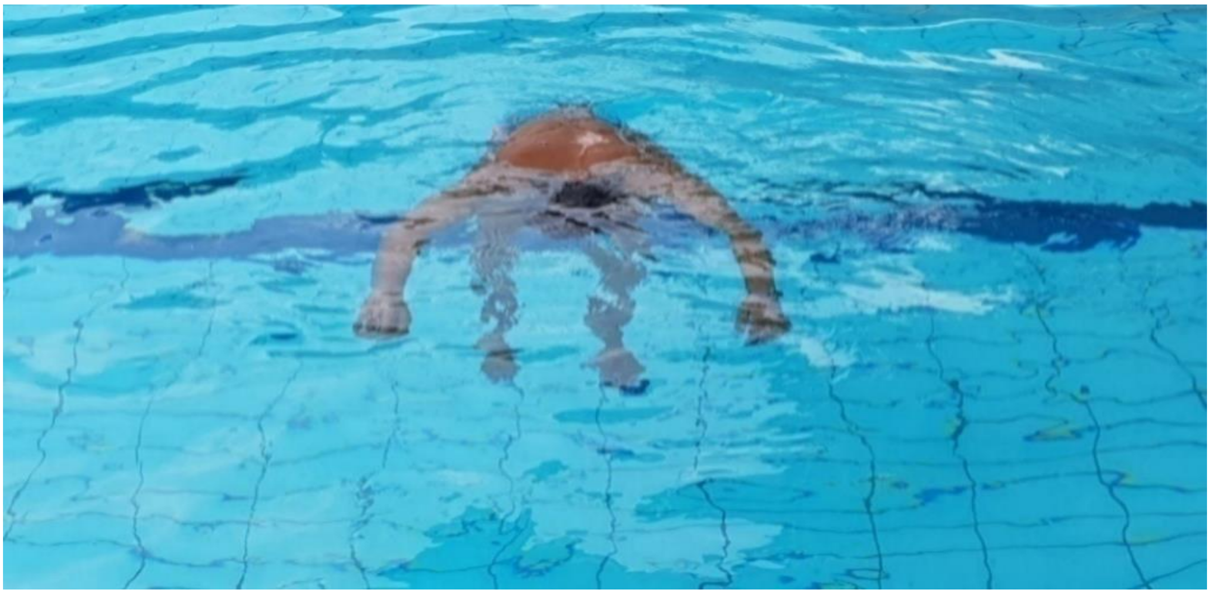


Figure G.2 – Passive or silent drowning

Figure G.3 – Swimmer in distress

Annex H
(informative)

Safety data sheet (SDS)

H.1 The SDS provides information about the hazards (e.g. physical, health, environmental, etc.) of a substance and how to use it safely (refer to SS 586-3). It also helps to identify, assess and control risks associated with the use of a substance in the workplace.

H.2 SDS of a chemical consists of the following 16 sections:

a) Section 1: Identification of the substance or mixture and of the supplier;

b) Section 2: Hazard identification;

c) Section 3: Composition/Information on ingredients;

d) Section 4: First-aid measures;

e) Section 5: Fire-fighting measures;

f) Section 6: Accidental release measures;

g) Section 7: Handling and storage;

h) Section 8: Exposure controls and personal protection;

i) Section 9: Physical and chemical properties;

j) Section 10: Stability and reactivity;

k) Section 11: Toxicological information;

l) Section 12: Ecological information;

m) Section 13: Disposal consideration;

n) Section 14: Transport information;

o) Section 15: Regulatory information; and

p) Section 16: Other information.

Annex I
(normative)

Hazardous chemicals

I.1 General

Hazardous chemicals are substances that have a health effect on people. The types of chemicals used to disinfect water within the aquatic industry usually fall into the category of hazardous chemicals.

For safety data sheets for the chemicals, hazard communication and provision of training for personnel to handle and manage the chemicals, SS 586-2 shall apply.

I.2 Safety signs for chemicals

Safety signs shall be installed around the swimming pool plant operations room, entrances and at storage areas. These signs are meant to alert emergency services to potential chemical hazards on the site and guide them in responding to chemical emergencies. For safety signs, refer to SS 508-5.

I.3 Safety measures for inlet valves

Inlet valves used for chemical loading shall be constructed to prevent accidental mixing of the chemicals during top-up, in accordance with SS 556.

Annex J
(normative)

Separation distances

J.1 Separation distances are determined by factors such as the types and quantities of the hazardous chemicals, their use, processes, and risk mitigation measures in the work area.

J.2 Separation can also be implemented by practising the following:

a) Never store different chemicals together;

b) Never mix chemicals;

c) Always wear the appropriate PPE;

d) Always keep liquids away from dry chemicals;

e) Always ventilate the storage areas;

f) Always check where gases accumulate; and

g) Always check the SDS for each chemical.

Bibliography

Standards/Publications

[1] SS 508-5 Graphical symbols – Safety colours and safety signs – Part 5: Registered safety signs

[2] SS 586-3 Specification for hazard communication for hazardous chemicals and dangerous goods – Part 3: Preparation of safety data sheets (SDS)

[3] Code of practice for water safety. National Water Safety Council

[4] Code of practice on environmental health. National Environment Agency Singapore, 2011

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Regulations/Acts

[1] Amusement Rides Safety Act 2011

[2] Building Control Act 1989

[3] Environmental Public Health Act 1987

[4] Fire Safety Act 1993

[5] Workplace Safety and Health Act 2006

[6] Workplace Safety and Health (First-aid) Regulations

[7] Workplace Safety and Health (Risk Management) Regulations

NOTE – The regulations and acts listed above are not exhaustive. Users of the standard will need to check with the relevant regulatory bodies on the latest regulatory and statutory requirements.

Resources/Websites

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NOTE – Hyperlinks are valid at the time of publication.

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To ensure adequate viewpoints are considered in the development and review of SSs and TRs, committees and working groups set up by the Standards Council consist of representatives from various key stakeholders which include industry associations, professional bodies, academia, government agencies and companies. SSs are also put up for public comment before publication.

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		Mr Koh Yeong Kheng	*Association of Small and Medium Enterprises of Singapore*
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KK Women's and Children's Hospital
MCST Association of Singapore
Ministry of Education
National University of Singapore
Ocean IFM Pte Ltd
Republic Polytechnic
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Sentosa Development Corporation
SGS Testing & Control Services Singapore Pte Ltd
Shangri-la Rasa Sentosa Resort, Singapore
Shangri-la Singapore
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Sport Singapore
Urgent Care Clinic International Pte Ltd

The Singapore Standards Council facilitates the development, promotion and review of standards for enterprise growth under the national standardisation programme overseen by Enterprise Singapore.

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